

MICHAEL P. RAGSDALE

Systems Architect | Enterprise Data Engineer Norfolk, VA (Willing to Relocate) | 757-289-1609 | hireme@michaelpragsdale.com | michaelpragsdale.com | github.com/raggiesoft

PROFESSIONAL SUMMARY

Systems Architect and Data Engineering Leader driven by a philosophy of "Empathetic Engineering"—the conviction that resilient data platforms must respect both the server's infrastructure and the end user's cognitive load. Proven track record operating as a technical "player-coach," designing complex ETL pipelines, bridging legacy databases with modern cloud environments, and establishing strict enterprise compliance frameworks. Specialist in leveraging open-source MVC architecture, secure cloud deployments, and production-grade AI/LLM integrations to drive institutional data strategies and accessible, zero-latency digital ecosystems.

TECHNICAL ARSENAL

- **Data Architecture & AI:** ETL / ELT Pipelines, Production LLM/AI Integration (Gemini), Relational & JSON Data Structures, Custom Open-Source MVC Frameworks
- **Languages & Scripting:** Python, PHP 8.4, SQL, JavaScript (ES6+), Bash / Shell, HTML5, CSS3/SCSS
- **Infrastructure & Cloud:** DigitalOcean (Droplets/Spaces), Nginx, Cloudflare Zero Trust (DNS/SSL), Decoupled CI/CD Pipelines
- **Compliance & Enterprise Governance:** Strict WCAG 2.1 AA, Section 508, ActiveNet CRM, JSON-LD / Schema.org, Deque axe

RELEVANT TECHNICAL EXPERIENCE

Principal Systems Architect (Open Source) RaggieSoft Media | Norfolk, VA | Est. 2008 – Present

- **Systems Architecture & Data Strategy:** Engineered and open-sourced the Stardust Engine CMS (MIT License), defining the long-term architectural roadmap for a custom PHP 8.4 MVC framework that serves massive-scale narrative data via a zero-latency file-based JSON architecture.

- **AI & LLM Integration:** Designed native API pipelines integrating advanced LLMs (Gemini) into the platform for automated data structuring, metadata generation, and narrative serialization within a governed digital framework.
- **DevOps & Cloud Leadership:** Directed a strategic infrastructure migration to high-performance DigitalOcean environments, utilizing Cloudflare edge networking, custom Nginx proxies, and decoupled Bash CI/CD pipelines (Jenna and Sarah) for automated, zero-downtime server synchronization.
- **Frontend Data Delivery:** Architected 'Elara', a custom Vanilla JS SPA router that parses complex JSON payloads to dynamically render accessible UI architectures, actively reducing cognitive load and embedding strict WCAG 2.1 AA/Section 508 compliance at the architectural root.

Clerk-Cashier (Functional Data Integration Lead) City of Virginia Beach Parks & Recreation | Virginia Beach, VA | May 2023 – Present

- **Data Engineering & ETL Pipelines:** Architect and manage manual ETL (Extract-Transform-Load) pipelines to ingest, normalize, and secure complex operational and financial data from legacy enterprise platforms (ActiveNet) into highly accessible digital formats.
- **Player-Coach Leadership & Training:** Serve as the primary technical trainer and operational escalation point for municipal staff, translating complex data integrity protocols and Revenue Cycle Management (RCM) compliance into actionable training programs.
- **Stakeholder & Governance Management:** Partner with facility leadership to optimize resource allocation, resolving complex logistical conflicts through data-driven insights and strict adherence to municipal policy.
- **Compliance Enforcement:** Govern output data to guarantee strict Section 508 compliance across all generated operational documents from legacy systems, ensuring institutional adherence to federal accessibility mandates.

Computer Operations Technician (Functional Accessibility Specialist) Old Dominion University (ODU Online) | Norfolk, VA | Nov 2019 – Jul 2022

- **Enterprise Accessibility Strategy:** Served as the primary technical gatekeeper for enterprise-level data and web content during a massive, university-wide digital infrastructure migration to the Canvas LMS.
- **Compliance Auditing:** Executed rigorous WCAG 2.1 AA and Section 508 compliance audits, remediating complex codebases (HTML/CSS) and ensuring strict structural integrity across institutional platforms using tools like Deque axe.
- **Architectural Standardization:** Engineered universal, inherently accessible data/architectural templates deployed across the university ecosystem, heavily streamlining data-entry workflows for instructional design faculty.

Mobile Applications Developer Intern Dominion Enterprises | Norfolk, VA | Sep 2012 – Dec 2012

- Engineered cross-platform mobile applications for iOS and Android utilizing the JavaScript-based Titanium SDK.
- Gained foundational, hands-on experience in mobile architecture and deployment, establishing a decade-long trajectory in software engineering.

EDUCATION & CERTIFICATIONS

- **Tidewater Community College** | Virginia Beach, VA Associate of Science, Information Technology (Focus: Software Engineering & Web Dev) | Expected Dec 2026
- **Old Dominion University** | Norfolk, VA Bachelor of Science, Customized Studies in Leadership | Prior Coursework
- **Wilson Workforce and Rehabilitation Center (formerly Woodrow Wilson Rehabilitation Center)** | Fishersville, VA Certificate, Business & Information Technology | 2011